CCNA 200-301 V2.0

13 CDP, LLDP and Documenting a Network

Part II — Switching and Network Access

EXAM TOPICS

2.3

LECTURER

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THE COURSE

Where this chapter sits

Part I	Foundations	chapters 1–7
Part II	**Switching and Network Access**	chapters 8–14
Part III	IP Routing	chapters 15–19
Part IV	Network Services and Security	chapters 20–26
Part V	Wireless, Virtualisation and Operations	chapters 27–32
Part VI	Sitting the Exam	chapters 33–34

What you will be able to do

- **Explain** what discovery protocols advertise and what they cannot tell you.
- **Configure** CDP and LLDP, including disabling them where they should not run.
- **Validate** a network diagram against the devices themselves, which is precisely what topic **2.3** asks.
- **Use** discovery output to find a mismatch — native VLAN, duplex or an undocumented device.

ASSUMED KNOWLEDGE

Before we start

Chapter 10 for native VLANs, Chapter 2 for duplex, Chapter 7 for filtering output.

KEY TERMS

Vocabulary for this chapter

- CDP
- LLDP
- LLDP-MED
- neighbour
- holdtime
- advertisement
- capability code
- platform
- local interface
- port ID

Each is defined where it first matters — not here.

ON THE EXAM

The verb here is validate

This is what the blueprint actually asks for

Topic **2.3** is “validate the accuracy of network documentation using CDP and LLDP”. It is not asking you to describe the protocols. It is asking you to take a diagram that claims `SW1 Gi0/1` connects to `R1 Gi0/0`, and prove from the devices whether that is true. Expect to be shown a diagram and some `show cdp neighbors` output that contradicts it.

DEFINITION

What a neighbour advertisement carries

Device identifier, the local and remote interface, the platform and capabilities, and depending on the protocol and platform: IP address, IOS version, native VLAN, duplex and power requirements.

Crucially, a neighbour is **directly connected at layer 2**. A device two switches away never appears. That is exactly what makes the output usable as evidence of physical cabling.

Reading the output

VERIFY

show cdp neighbors

SW1#

```
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone

Device ID    Local Intrfce   Holdtme   Capability   Platform    Port ID
R1           Gig 0/1          142        R S I      C1111       Gig 0/0/0
SW2          Gig 0/2          167        S I        WS-C2960    Gig 0/1
SW3          Gig 0/3          121        S I        WS-C2960    Gig 0/24
AP-EAST      Gig 0/10         155        H          AIR-AP2802  Gig 0
```

VERIFY

show cdp neighbors detail

SW1#

```
-------------------------
Device ID: R1
Entry address(es):
  IP address: 10.0.0.1
Platform: Cisco C1111-4P,  Capabilities: Router Switch IGMP
Interface: GigabitEthernet0/1,  Port ID (outgoing port): GigabitEthernet0/0/0
Holdtime : 142 sec

Version :
Cisco IOS XE Software, Version 17.09.04a

Duplex: full
Native VLAN: 999
```

CONFIGURE

Running both protocols, and switching them off where they do not belong

IOS · configuration mode

```
! CDP is on by default; LLDP is not.
lldp run
!
! Turn CDP off on a port facing anything you do not control.
! Advertising your platform and IOS version to a guest network,
! an ISP handoff or an untrusted uplink is free reconnaissance.
interface GigabitEthernet0/24
 description ISP handoff - no discovery
 no cdp enable
 no lldp transmit
 no lldp receive
!
! Globally, if policy demands it:
!   no cdp run
```

WATCH OUT

Do not disable it everywhere on principle

There is a reflex to switch CDP off across the estate “for security”. Weigh it properly: internally, CDP and LLDP are how you validate cabling, how phones learn their voice VLAN (Chapter 9), and how native VLAN and duplex mismatches are detected. Disable them on untrusted edges — ISP links, guest ports, DMZ segments — and leave them running inside.

Validating a diagram

WORKED EXAMPLE

The diagram says one thing, the switch says another

The documentation claims:

- `SW1 Gi0/2` → `SW2 Gi0/1`
- `SW1 Gi0/3` → `SW3 Gi0/1`

The output above shows `Gi0/2` to `SW2 Gi0/1` — correct — but `Gi0/3` to `SW3` **`Gig 0/24`**, not `Gi0/1`.

What that means in practice. Somebody re-patched `SW3` and did not update the diagram. It may be harmless. It may not: if `SW3 Gi0/24` is configured as an access port while `Gi0/1` was the trunk, the uplink is carrying one VLAN instead of all of them, and the fault will surface later as “the new VLAN does not work on the third floor”.

What to do. Correct the diagram, then check that the port actually in use carries the configuration the design intended. Discovery output tells you what is plugged in; it does not tell you whether the port is configured correctly. That is a second check.

Common pitfalls

- **Expecting to see non-adjacent devices.** Neighbours are one hop. A device that does not appear may simply be further away.
- **Concluding a device is absent when CDP is off.** A non-Cisco switch, or one with CDP disabled, is invisible to CDP and may still be there. Cross-check with LLDP and the MAC address table.
- **Trusting stale entries.** Holdtime counts down from 180 s. An entry can persist for three minutes after a device is unplugged.
- **Forgetting `lldp run`.** Without it, `show lldp neighbors` is empty and you may wrongly conclude nothing is connected.

Common pitfalls (continued)

- **Reading the port columns backwards.** `Local Intrfce` is yours; `Port ID` is theirs.

An undocumented switch appears in the discovery table.

An audit of `SW1` shows a neighbour that is not on any diagram:

An undocumented switch appears in the discovery table.

What would you check first — and what do you expect to see?

show cdp neighbors GigabitEthernet0/14 detail

SW1#

```
-------------------------
Device ID: Switch
Entry address(es):
Platform: cisco WS-C2960-8TC-L,  Capabilities: Switch IGMP
Interface: GigabitEthernet0/14,  Port ID (outgoing port): FastEthernet0/1
Holdtime : 131 sec

Version :
Cisco IOS Software, C2960 Software, Version 12.2(55)SE
```

What is the fault — and what does this output rule out?

Why it is doing that

Diagnosis. An eight-port switch with the default hostname `Switch`, no management address, and a decade-old IOS version, connected to an access port — somebody has plugged a desk switch into a wall socket to get more ports. It is unmanaged, it participates in spanning tree, and it can influence the root election (Chapter 12).

And the lesson

Fix. Short term: confirm what is downstream of it before unplugging anything, because there are probably users behind it. Long term, this is exactly what `spanning-tree bpduguard` on access ports is for — with it configured, the port would have errdisabled the moment the switch was connected and the problem would have announced itself instead of being found in an audit months later.

Draw the network from the devices

Topology. Four switches and a router, cabled by your instructor or a partner without telling you how, plus a deliberately wrong diagram.

1. Using only `show cdp neighbors` from each device, draw the physical topology. Do not look at the cables.
2. Compare your drawing with the supplied diagram and list every discrepancy.
3. Run `show cdp neighbors detail` on one link and record the IOS version, duplex and native VLAN of the neighbour.
4. Create a native VLAN mismatch on one trunk and find the resulting log message.
5. Enable `lldp run` everywhere and compare `show lldp neighbors` with the CDP view. Account for any difference.
6. Disable CDP on one port and confirm the neighbour disappears — noting how long it takes.

Draw the network from the devices (continued)

7. **Extension.** Write a short procedure a junior technician could follow to validate a patching change using discovery output alone.

CHECKPOINT 1 of 3

`show cdp neighbors` lists a device you know is three switches away. What must be true?

Discuss · then advance

ANSWER

Checkpoint 1 of 3

`show cdp neighbors` lists a device you know is three switches away. What must be true?

> There is no switch between you and it that processes CDP — so the intervening devices are hubs, media converters, or unmanaged switches that pass the frames through. Neighbour discovery only shows *directly connected* devices, so a distant one appearing means the path is not what the diagram says.

CHECKPOINT 2 of 3

You unplug a device and it still appears in the table. Why, and for how long?

Discuss · then advance

ANSWER

Checkpoint 2 of 3

You unplug a device and it still appears in the table. Why, and for how long?

> Because entries persist for the advertised holdtime, 180 seconds by default for CDP. Until it expires, the table describes a neighbour that is no longer there.

CHECKPOINT 3 of 3

Which single command would tell you a neighbour’s IOS version and native VLAN?

Discuss · then advance

Checkpoint 3 of 3

Which single command would tell you a neighbour’s IOS version and native VLAN?

> `show cdp neighbors detail`. The summary form gives neither.

What to take away

- CDP is Cisco and on by default; LLDP is open and off by default. Run both in a mixed estate.
- Neighbours are directly connected only — which is what makes the output evidence of cabling.
- `Local Intrfce` is your port, `Port ID` is theirs. Those two columns plus the device name are the diagram.
- `detail` adds IP, version, duplex and native VLAN, and catches mismatches before they cause incidents.

What to take away (continued)

- Disable discovery on untrusted edges, not everywhere. Holdtime means entries persist for minutes after a device is removed.

Where we got to

- CDP is Cisco and on by default; LLDP is open and off by default. Run both in a mixed estate.
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- `Local Intrfce` is your port, `Port ID` is theirs. Those two columns plus the device name are the diagram.
- `detail` adds IP, version, duplex and native VLAN, and catches mismatches before they cause incidents.

NEXT

Chapter 14 — Troubleshooting Layer 2